Piter Z. Garcia Bautista

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Research Profile

Graduate researcher with a strong foundation in machine learning methods, statistical learning theory, and data-driven decision support. Background includes fairness-aware ML for high-stakes settings, reproducible evaluation frameworks, bandit and sequential decision-making algorithms, and applied ML across healthcare, bioinformatics, and network routing domains. Current Graduate Assistant work (Quantum and AI Research, RIT) emphasizes principled model evaluation, statistical rigor, and method reliability across heterogeneous and partially observed data environments. Interested in pushing ML methodology forward — building models that generalize robustly and whose failure modes are understood and bounded.

Research Interests

Machine learning theory and methods; statistical learning and model evaluation; fairness-aware and robust ML; bandit and sequential decision-making; reproducible benchmarking; applied ML in healthcare, bioinformatics, and network systems.

Education

Rochester Institute of Technology — Expected Aug 2026
Master of Science, Data Science — GPA 3.9 — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Data-Driven Knowledge Discovery (ISTE-780), Neural Networks & Deep Learning, Applied Data Science (DSCI-601), Bioinformatics Algorithms (BIOL-630), Software Engineering for Data Science (DSCI-644), Foundations of Data Science (DSCI-633).

University of Rochester, Warner School of Education — In progress
Graduate study, Teaching Computer Science K–12 (M.S. track) — GPA 4.0 — *Rochester, NY*

- NSF Robert Noyce Scholar ($30,000 full funding); Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology — 2015
Master of Science, Computer Science — GPA 3.2 — *Rochester, NY*

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.
- Graduate research: *Big Data Medical Diagnosis: A Multi-Method Approach* — ML ensemble methods for clinical classification.

Farmingdale State College — 2012
Bachelor of Science, Computer Engineering — GPA 3.3 — *Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.

- Honors: CSTEP Award & Merit; Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology, Software Engineering Department Fall 2025 – Present
Graduate Assistant, Quantum and AI Research *Rochester, NY*

- Build reproducible ML evaluation pipelines across local, Colab, and GCP environments with structured logging, hyperparameter tracking, and statistical validation.
- Develop bandit and sequential decision-making systems (UCB, neural-UCB, contextual bandits) with rigorous evaluation frameworks comparing algorithmic variants under controlled conditions.
- Produce manuscript-ready technical analyses on ML method performance, including evaluation under distribution shift, partial observability, and adversarial settings.
- Design fairness-aware evaluation protocols that measure both predictive performance and equity metrics across heterogeneous subpopulations.

Rochester Institute of Technology 2024 – 2025
Equitable ML Research — ISTE-780 Project (Data-Driven Knowledge Discovery) *Rochester, NY*

- Pioneered first systematic application of fairness assessment to RNA secondary structure prediction algorithms (EQUITAS framework), revealing critical equity violations (11.1% equity score) across demographic proxies.
- Developed modular ML evaluation codebase (900+ lines) implementing reproducible fairness-aware algorithm comparison with automated equity-metric reporting.
- Achieved statistical significance ($p < 0.01$) in bias detection across multiple algorithmic variants using SHAP and disparity-impact analysis.

Rochester Institute of Technology 2025 – 2026
Computational Biology Research (BIO614 and BIO550) *Rochester, NY*

- Applied ML and statistical methods to RNA structure prediction, including evaluation of probabilistic models against classical dynamic-programming baselines.
- Conducted RNA-seq differential-expression analysis with reproducible preprocessing, QC validation, and statistical-significance reporting.

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed scalable Python and pandas ML workflows for ingestion, feature engineering, and statistical diagnostics in production environments.
- Implemented structured quality checks and anomaly-detection pipelines strengthening production data reliability.

VIOME Inc. Jan 2021 – Sep 2022
AI Data Solutions Engineer *New York, NY*

- Developed healthcare ML workflows for feature engineering, model experimentation, and evaluation — predictive health applications in AWS production environments.
- Built reproducible model-evaluation infrastructure with monitoring components to detect performance degradation under distribution shift.

Selected Technical Writing

- *Quantum Path Optimization with Fault-Tolerant Routing* — manuscript-level analysis of quantum-classical hybrid ML for fault-tolerant routing under noise constraints (under review, 2025–2026).
- *Fairness-Aware Bandits for Network Routing in Quantum and Clinical Settings* (DSCI601 project, manuscript-style) — bandit learning with equity constraints for high-stakes sequential decisions.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report) — systematic fairness assessment applied to ML-driven diagnostics.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report) — ensemble and regularized regression methods for clinical risk stratification.
- *BIO614-FinalProjectProposal* (Overleaf) — formal computational methods manuscript with rigorous benchmarking and reproducible evaluation.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost; bandit/RL (UCB, contextual bandits, Thompson sampling); fairness-aware ML; model evaluation and uncertainty quantification; SHAP analysis
Data and Statistics: pandas, NumPy, SciPy, statistical modeling, hypothesis testing, reproducible benchmarking
Infrastructure: Git/GitHub, Docker, Linux, GCP, AWS, LaTeX

Selected Fit for KTH Machine Learning Position

- Strong theoretical and applied ML background spanning supervised, sequential decision-making, and fairness-aware methods, with emphasis on rigorous evaluation methodology.
- Hands-on experience designing and running controlled ML experiments with statistical validation — directly applicable to research that advances ML methodology.
- Demonstrated commitment to reproducibility: all research projects include structured evaluation pipelines, logging, and statistical-significance reporting.
- Deep interest in understanding when and why ML methods work or fail, with particular focus on robustness, equity, and reliability in high-stakes applications.